

Pitch and Harmonics

In the previous note, we saw that a vibrating string clamped at both ends can only vibrate at a certain set of frequencies. Let the string have length L and the speed of propagation in the string be $v = \sqrt{\frac{T}{\lambda_m}}$ where T is the tension in the string and λ_m is the mass per unit length of the string. Then the string can oscillate with wavelength λ (not the same as λ_m) for any wavelengths that have nulls at the ends of the string. This happens whenever we fit a multiple of *half* a wavelength onto the string since a sine wave is 0 at the ends and the middle of a period. The lowest frequency the string can oscillate at is $\nu_0 = v/2L$ where we call ν_0 the *fundamental*. The string can oscillate at any multiple of the fundamental: ν_0 , $2\nu_0$, $3\nu_0$ etc. A string can oscillate at any of these frequencies (or any combination of them), and your ear will hear each one as a different pitch. These different frequencies are called *harmonics* or *overtones*.

1. Pitch

Our ear is a logarithmic detector. When it comes to pitch, that means that the difference in pitch you hear is set by the *ratio* of the frequencies. The most fundamental interval in music is the octave; your ear hears notes separated by an octave as the same pitch but higher/lower. When you go and measure the frequencies of notes separated by an octave, what you find is that they are exactly a factor of 2 different in frequency. If you go two octaves up, you double the frequency to go up the first octave, then double that to go up the second octave, so the frequency of the upper note is $2 \times 2 = 4\nu_0$ where ν_0 is the lower frequency. If we want to go up three octaves, our new frequency is $2 \times 2 \times 2 = 8\nu_0$.

What will our ears hear for the different harmonics? The lowest two frequencies are ν_0 and $2\nu_0$ - these are an octave apart. The second and the fourth are $2\nu_0$ and $4\nu_0$ - once again, separated by an octave. What about $3\nu_0$? Your ear hears that as a perfect fifth above $2\nu_0$. If our fundamental is a C, then the first overtone is the C above that, and the next is the G above that. Going from that G to the next C (which is $4\nu_0$) gives us a fourth. The next intervals are a major third ($4\nu_0$ to $5\nu_0$) and then a minor third ($5\nu_0$ to $6\nu_0$). Even though the harmonics are equally spaced in frequency, your ear hears them get closer and closer together.

2. Timbre

Our string can oscillate at any of the allowed frequencies. Since the sum of any two solutions to the wave equation is also a solution, our string can also oscillate with *multiple* frequencies. In fact, this is what usually happens. When you bow a violin string, on a microscopic scale the bow is randomly tugging on the string. That randomness in turn means that the string is oscillating at all the harmonic frequencies. You might think that this would lead us to hear several notes at the same time, but that's not what happens. Instead, your ear tends to hear all the harmonics as a single note. The relative strengths of the different harmonics sets the sound quality or *timbre* of a note. Different instruments excited different harmonics with different strengths, and that's a huge

part of how you tell the difference between a flute and a french horn.

If you want to play with this (and can run python example codes), look at `play_harmonics.py` on github. The vector “amps” sets the amplitude of each harmonic. If you want to have the first three harmonics (so, say C3, C4, and G4) then you would set “amps=[1,1,1]”. Play around and see if you can reproduce an instrument! Generally, the more power in the higher harmonics there is, the brighter the sound quality. Most of the power in the lower harmonics leads to a warmer sound. If you pluck a violin string, where along the string you pluck changes the pattern of the harmonics, which leads to different sounds. Some instruments only have odd harmonics - your ear hears a very different sound quality from those.

Your brain also combines signals in non-obvious ways. The lowest note a cello can play is the C two octaves below middle C, or about 65 Hz. We’ll see later in the course that a cello is not actually big enough to play a note that low. It *can* play the harmonics at 130 and 195 Hz. Your brain hears those and say “OK, I’m hearing all the harmonics of 65 Hz, so that note must be 65 Hz.” You *hear* a note at 65 Hz because that’s how the brain can make sense of what it’s hearing.

3. Scales and the Circle of Fifths

Western music breaks up an octave into 12 notes, and the harmony is based on intervals with simple ratios (3:2 fifths, 4:3 fourths *etc.*). Harmonics let us understand why this is. If I play two notes with a ratio of 3:2 then you get what we generally interpret as a pleasing sound. It’s natural for instruments to produce sounds with these ratios. If I tweak that ratio slightly, then we’ll hear beating which humans generally find unpleasant. We can also get beating from harmonics. Let’s say we have two violins playing a fifth apart. The first one will have signals at $\nu_0, 2\nu_0, 3\nu_0$ and so on. The second violin will have signals at $1.5\nu_0, 3\nu_0, 4.5\nu_0, 6\nu_0$ and so on. Both instruments have harmonics at $3\nu_0$ and $6\nu_0$ (and so on). If our violinists play perfect fifths, those harmonics will reinforce each other. If they’re out of tune, though, then the harmonics will beat and our ear will not like that.

The 3:2 ratio also sets what we call *the circle of fifths*. If you go up a fifth from a C you get a G. If you do it again, you get a D, and then an A, and E and so on. After you do this 12 times, you should end up back where you started at. Mathematically, we’re saying that $1.5 \times 1.5 \times 1.5 \dots$ of 1.5^{12} should be a power of 2. If we’re careful with counting, we can even say *which* power of 2. One loop around the circle of fifths brings you up by 7 octaves (since there are 7 half-steps in a fifth). If you plug in 1.5^{12} , you see that the value is 129.746, which is close to the value of $2^7 = 128$ that we expected. It isn’t *exactly* right, though. We’re off by 1.3%. That’s far enough off that your ear absolutely hears the interval is wrong.

Figuring out how to handle the error in the circle of fifths was an important challenge for the musical community hundreds of years ago. The simplest thing to do if you were say building an organ was to tune the fifths to be exactly in the 3:2 ratio, and just leave one interval (usually B to F#) off by 1.3%. If you were playing in C major, this worked great! All of your intervals were ratios of small integers, and your chords didn’t beat. However, if you tried to write a piece in say B major, then a major chord would sound just awful because the B and F# were out of tune. The

beating pattern reminded musicians of the howling of a wolf, so the out-of-tune chord is known as the “wolf fifth.”

Today, we usually break up the octave into 12 equal parts. If the ratio between half-steps is r , then the ratio of C# to C should be r . The ratio of D to C# should again be r , so the ratio of D to C should be $r * r = r^2$. Since we have 12 half-steps in an octave (because the circle of fifths takes 12 steps to bring you back almost to where you started), the ratio of two notes an octave apart should be $r * r * r... * r = r^{12}$. We know that ratio should be 2, so we get that $r^{12} = 2$ or $r = 2^{1/12} = 1.05946...$ This is called *equal tempering* and we can play a piece in any key, because no intervals are especially bad. One JS Bach was so happy about equal tempering that he wrote not one but two books for the keyboard with a prelude and fugue in each possible key (major and minor for each half-step in the scale), *The Well-Tempered Klavier*.

Equal tempering is not perfect, though. In an equal tempered system, the frequency ratio for a fifth is $2^{7/12} = 1.4983$. This is very close to 1.5, but is still about 0.11% off. The thirds are worse - a major third should have a ratio of 5:4, but in equal tempering it's almost exactly 1.26 - an error of 0.8%. A minor third should be 6:5 but it's even worse than the major third, with an error of 0.9%. These differences are on the edge of what people can perceive (there's a wide range in sensitivity to relative pitch). We'll listen to some of these in class and see what you think!